AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 29 · STUDENT LESSON

Find a Real Problem

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Name the need. Define the outcome. Keep the solution open.

Start the final fictional activity project, inspect limited observations and write a bounded brief before choosing a mechanism.

Keep one habit: Read the evidence → define useful outcomes → compare a simple alternative.

MAIN LESSON · SECTION 1

What problem would this project actually solve?

> Begin with observed difficulty rather than a preferred technology.

Start before the solution

“Build an AI app” names a possible solution. A useful project brief first names who needs help, what happened and how you will recognise improvement.

Recall the limits of a paper-project permission

A teacher permits a paper prototype using fictional activity requests. Does this permit collecting real pupils' names and publishing their choices? Explain.

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Your learning goals

- I can describe a specific **user** difficulty using supplied **evidence** without inventing its cause.
- I can define a useful **outcome** and a way to observe it before choosing a solution.
- I can choose a small project with clear boundaries, permitted information and a simple alternative.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

A strong project begins with a specific **user** and observable **need**.
</details>



LEARNING TARGET Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can turn a recorded problem into a bounded, measurable project goal.

Begin the final paper activity-recommender sequence

> Use the supplied fictional log; a recommendation remains a choice.

Meet the users and project sequence

For the final project sequence, we will develop a small activity recommender on paper. Its **user**s are fictional pupils choosing a classroom activity that fits their available time and room conditions.

We will decide what the recommender must do, build a traceable **process**, test it and explain its limits. A recommendation is a suggestion; the pupil can decline it.

Read the ten-request observation

Start before the solution. Our supplied **observation** log describes ten fictional requests made using an activity catalogue. Six first choices fit the request's time and room conditions. Four did not and **need**ed another choice.

These invented classroom records give us a problem to investigate. They are not **observation**s of your classmates and do not show how often this happens in other classes.

Write only what the evidence establishes

A useful **problem statement** is: "In the supplied ten-request exercise, four first choices did not fit the stated time or room conditions, so those **user**s had to choose again." "We **need** AI" names a possible technology but says neither who **need**s help nor what should improve.

The log also does not establish whether unclear wording, missing information or another cause produced the mismatches.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY 'Students **need** AI' is not a **problem statement**; 'students spend five minutes finding misplaced supplies after art' is observable and testable.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

MAIN LESSON · SECTION 3

Define useful outcomes before building

> Check compatibility and honest no-match results without promising enjoyment.

Separate observation, need and project boundary

From an observed difficulty to a testable goal

EVIDENCE OR DECISION	WHAT IT SUPPORTS	WHAT REMAINS TO CHECK
4 of 10 choices mismatch	Some re-selection occurred	Cause is not established
Need a compatible choice	Check time and conditions	User can still decline
Paper suggestion process	Small permitted prototype	Benefit needs a comparison
Measure fit and coverage	Observe useful outcomes	Do not hide no-result cases

The goal describes a useful result. Building a tool is an activity, not evidence that users benefited.

Define compatibility and no-match success

Our target **outcome** is a compatible suggestion when the catalogue contains one, and a clear no-match message when none fits.

Compatibility means respecting the stated conditions; it does not mean that every pupil will enjoy the activity. We can check time and room conditions from a request and catalogue. Enjoyment would **need** different **evidence**.

Count helpful results as well as wrong suggestions

- Count compatible suggestions, incompatible suggestions and no-match messages separately. Otherwise a system could avoid every wrong suggestion by refusing to suggest anything.

  If a valid option existed, that refusal would still fail to provide the intended help. Record the total number of requests so a count has a clear denominator.

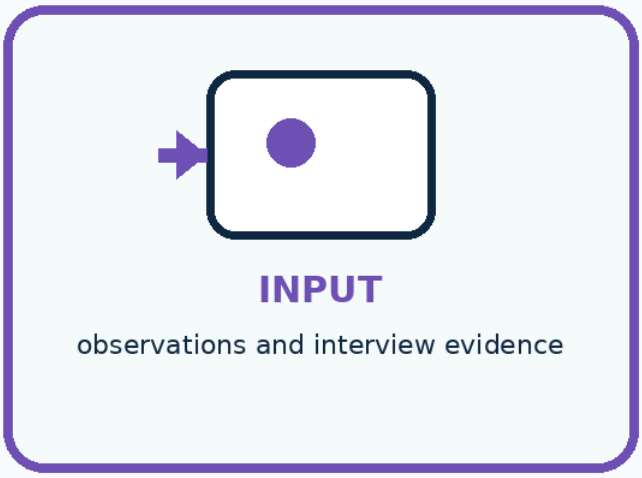
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

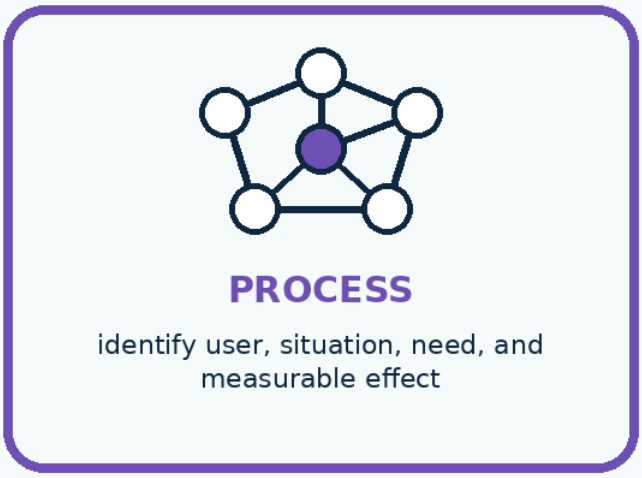
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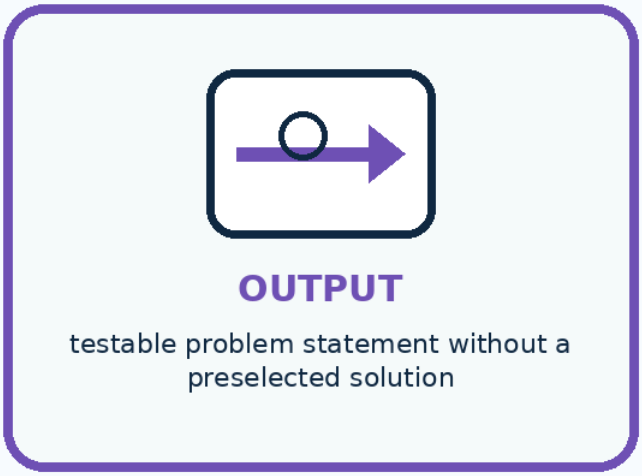
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT observations and **interview evidence**

PROCESS identify **user**, situation, **need**, and measurable effect

OUTPUT testable **problem statement** without a preselected solution

CORE IDEA A strong project begins with a specific **user** and observable **need**.

UNDER THE HOOD Problem framing separates **evidence** about a **user**'s **need** from assumptions about a solution. A strong statement identifies who, when, what observable difficulty occurs, and how improvement could be measured.

Draw a fresh example with all three stages:

A closer explanation

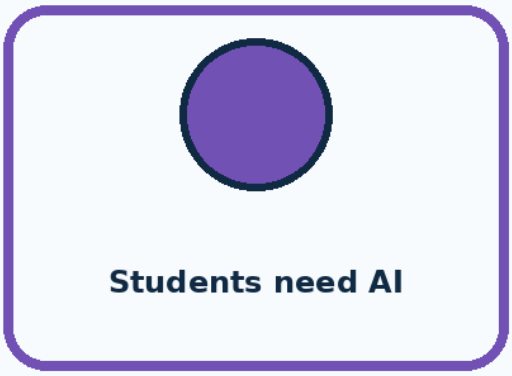
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare technology-first claims with observable needs

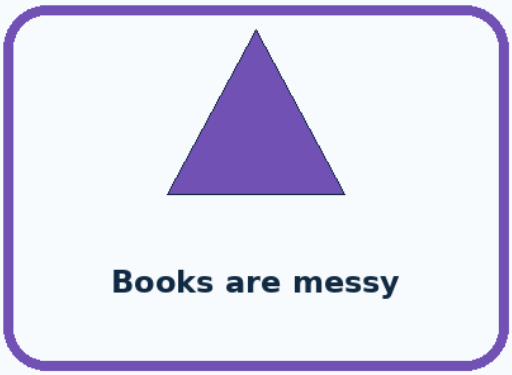
> A specific burden and a limited observation suggest questions to test.

Pictures paired with their evidence



The case to investigate

CASE 1 — Students **need** AI Clue: technology-first claim. Meaning: weak framing.



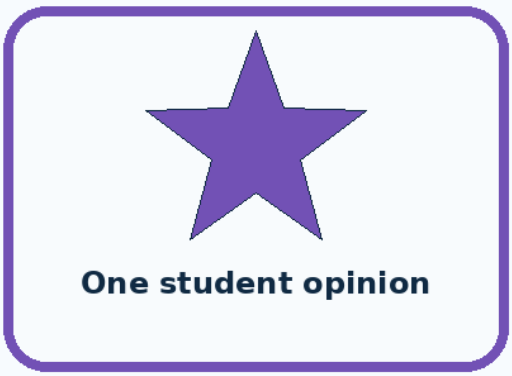
The case to investigate

CASE 2 — Books are messy Clue: vague judgment. Meaning: **need**s **observation**.



The case to investigate

CASE 3 — Librarian re-sorts 20 books daily Clue: specific burden. Meaning: testable **need**.



The case to investigate

CASE 4 — One student opinion Clue: useful but limited. Meaning: seek more **evidence**.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

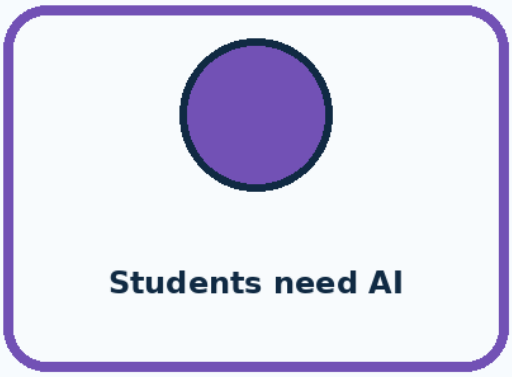
BOUNDARY Do not collect personal stories unrelated to the design question.

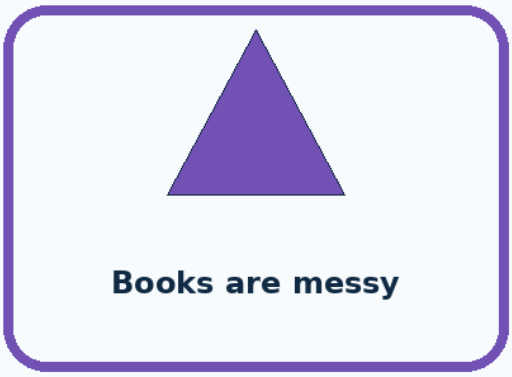
▼ Earlier wording, alternatives and picture transcripts

PICTURE GALLERY

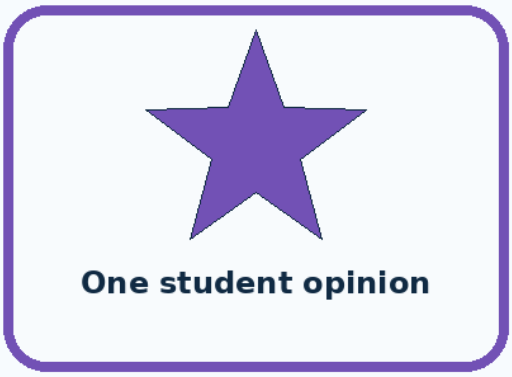
Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Students **need** AI Clue: technology-first claim. Meaning: weak framing.

CASE 2 — Books are messy Clue: vague judgment. Meaning: **need**s **observation**.

CASE 3 — Librarian re-sorts 20 books daily Clue: specific burden. Meaning: testable **need**.

CASE 4 — One student opinion Clue: useful but limited. Meaning: seek more **evidence**.

Change one thing in the pictured case

Change: The approved list contains no tray with every requested tool.

Prediction: The finder should report no suitable tray rather than guess.

Reason: Success includes honestly identifying unsatisfied requirements.

Trace the separate craft-tray problem

> Keep the eight tray requests separate from the ten activity requests.

Read the tray observation and permitted scope

A fictional class loses time finding the correct craft tray.

In eight recorded requests, three selected trays lack a required tool; five match.

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.

Return to the ten-request activity problem

AI ACADEMY / AI-1/29 · 03 · INSPECT THE LESSON MODEL

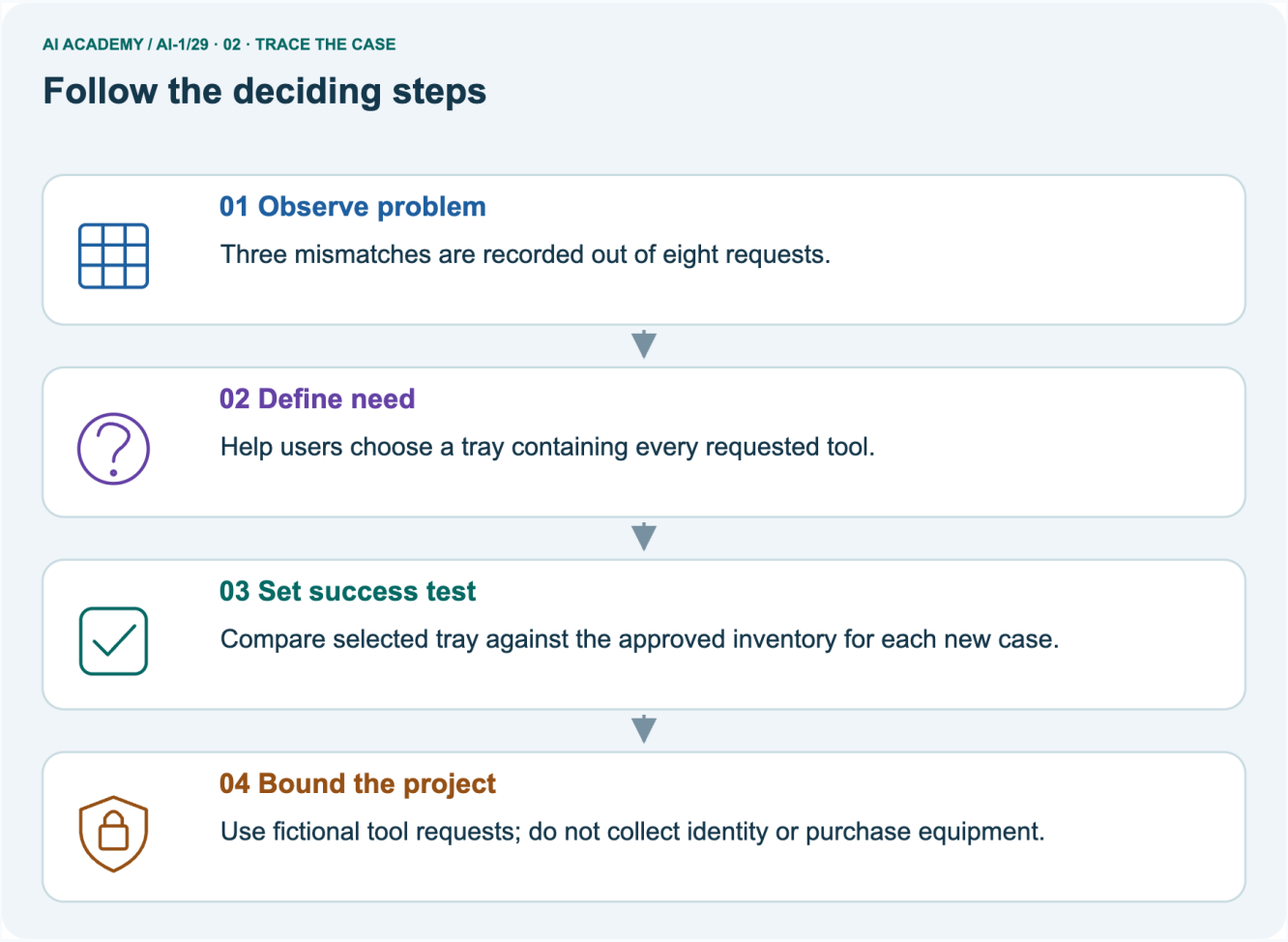
From an observed difficulty to a testable goal

Evidence or decision	What it supports	What remains to check
4 of 10 choices mismatch	Some re-selection occurred	Cause is not established
Need a compatible choice	Check time and conditions	User can still decline
Paper suggestion process	Small permitted prototype	Benefit needs a comparison
Measure fit and coverage	Observe useful outcomes	Do not hide no-result cases

The goal describes a useful result. Building a tool is an activity, not evidence that users benefited.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace observation, goal and success check



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Observe problem	Three mismatches are recorded out of eight requests.
2. Define need	Help users choose a tray containing every requested tool.
3. Set success test	Compare selected tray against the approved inventory for each new case.
4. Bound the project	Use fictional tool requests; do not collect identity or purchase equipment.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Explain the measurable goal

- The mismatch record describes a concrete **need**. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

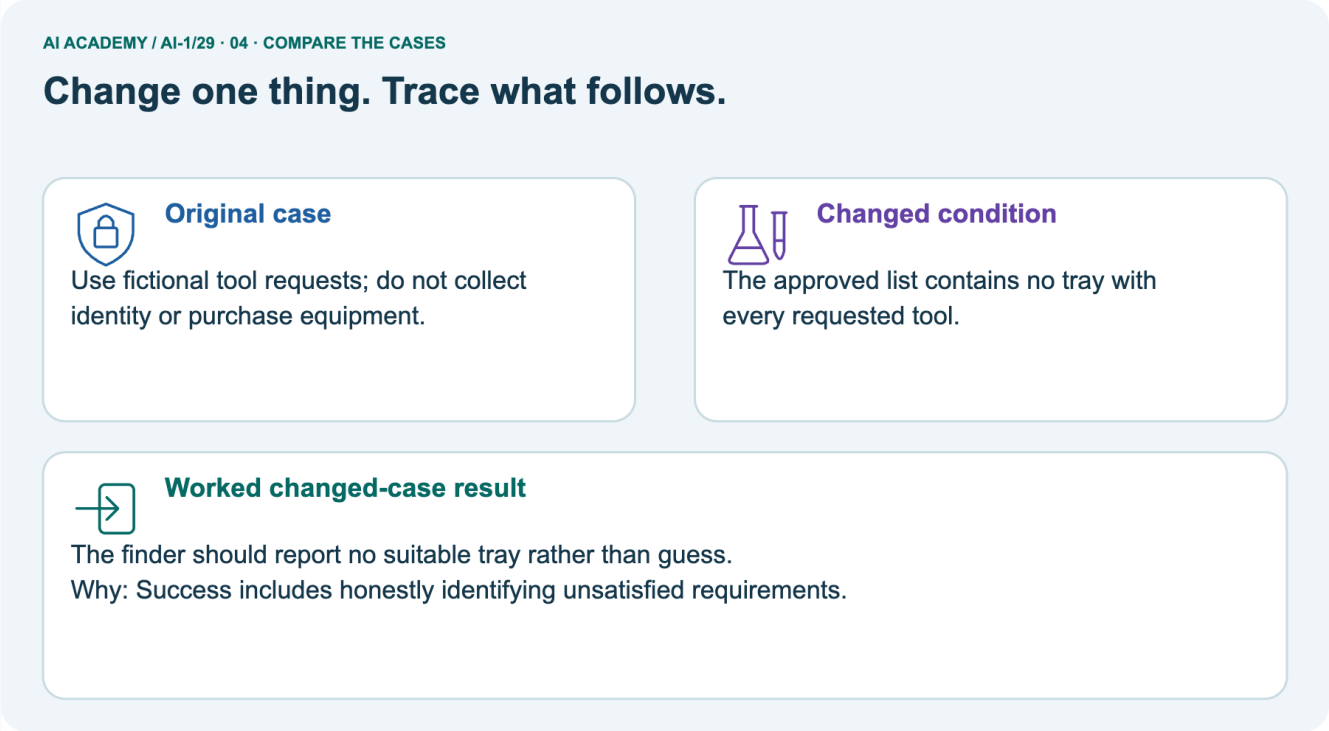
  Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

- Read the labelled picture: Write a one-sentence project goal and say what **evidence** would count as a failure.

Predict when no tray satisfies the request

- Before opening the no-suitable-tray picture, suppose no approved tray contains every requested tool. What result would respect the requirement? How will you distinguish this genuine no-match case from a missed valid option?

Check the no-match response



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit the population and technology claims

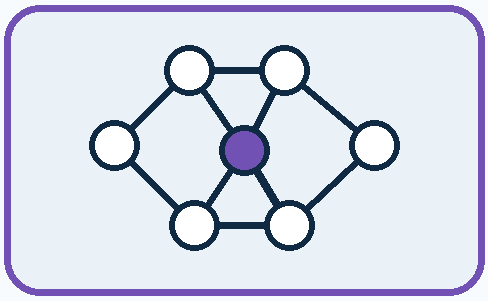
Eight requests do not show the problem's frequency across the entire school or prove AI is necessary.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE 'Students **need** AI' is not a **problem statement**; 'students spend five minutes finding misplaced supplies after art' is observable and testable.

STEP 1 — OBSERVE **observation**s and **interview evidence**

STEP 2 — TRACE identify **user**, situation, **need**, and measurable effect

STEP 3 — CONCLUDE testable **problem statement** without a preselected solution

MODEL REASONING A strong statement names the **user**, context, observable difficulty, and desired improvement without requiring AI.

Protect privacy, avoid leading questions, and distinguish what the **user** said from your inference.

DO NOT OVERCLAIM Do not collect personal stories unrelated to the design question.

What new **evidence** could change this answer?

A closer explanation

Project choice	Boundary for this sequence	Reason
Users and setting	Fictional classroom activity requests	A clear situation to test

Project choice	Boundary for this sequence	Reason
Information	Available minutes and quiet-room requirement	Relevant to compatibility
Action	Suggest an activity or report no match	The user retains the choice
Outside the project	Names, profiles, purchases and public posting	Not needed for this paper task

The teacher permits a paper prototype using fictional requests and an approved catalogue. We do not **need** names, home details or a history of a pupil's choices.

A request can say "10 minutes, quiet room required" without identifying anyone. Use only the request information **need**ed to test the intended **outcome**.

A clear catalogue or a short checklist may solve this problem. We will compare a proposed recommender with a simple starting method instead of assuming a more complicated system is better.

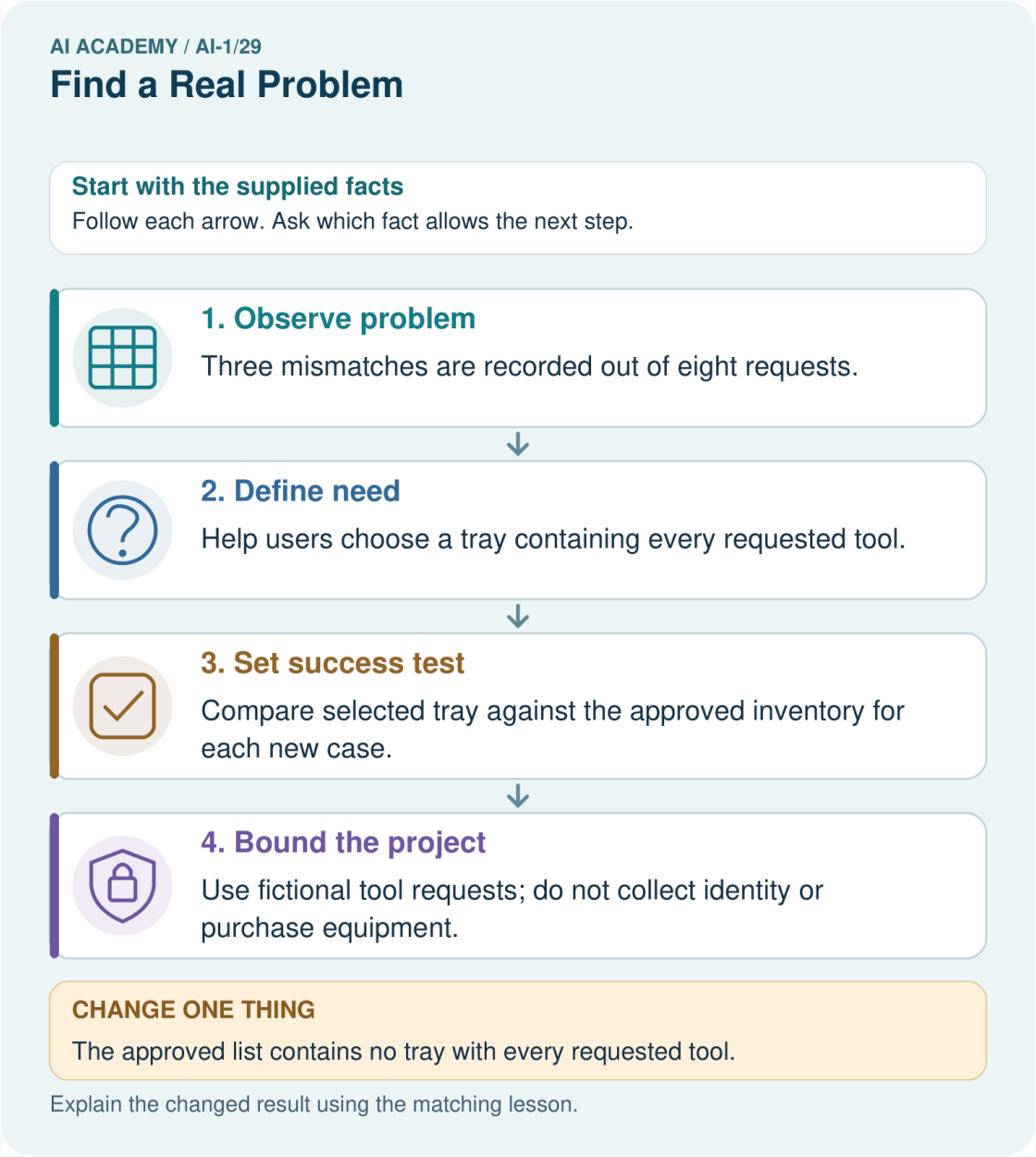
This project can use explicit rules; it does not have to fit a machine-learning model. The choice should follow the problem, available **evidence** and limits of the task.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Find a Real Problem

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing: The approved list contains no tray with every requested tool. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Specify what would count against your solution

> Keep motivation records separate from evaluation of a tool built later.

Compare against a useful simple method

Before building, say what would count against your idea. If a clear catalogue already produces compatible choices for all supplied cases, adding a recommender might offer little benefit on those cases.

If the recommender hides valid options or produces unexplained no-match messages, revise it or keep the simpler method. A project can succeed as an investigation even when the **evidence** favours a simpler solution.

Label observation, plan and evaluation separately

Keep the ten-request **observation** log separate from later evaluation records. It motivates the problem; it is not a fair test result for a tool that has not been built.

Later weeks will introduce explicit requirements and test cases, including a reserved final set. Label plans as plans and measured results as results.

Repair both tempting claims

- CLAIM A A popular technology is proof of a real problem.
- CLAIM B The first **interview** represents every **user**.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Students need AI	technology-first claim	weak framing
Books are messy	vague judgment	needs observation
Librarian re-sorts 20 books daily	specific burden	testable need
One student opinion	useful but limited	seek more evidence

Repair two tempting claims

Repair:

Repair:

A closer explanation

> **A proposed system always says “Ask the teacher”. It never suggests an incompatible activity. Has this alone demonstrated the intended benefit? Explain which additional result you would need to check**

MAIN LESSON · SECTION 8

Practise: separate findings from hypotheses

> A mismatch count does not establish everyone’s feelings or its cause.

Repair the unsupported catalogue story

Suppose a team writes “All pupils hate the catalogue because it is too long.” The supplied log says nothing about every pupil’s feelings or the cause of the mismatches.

A more careful note is “Four of ten first choices were incompatible; the cause is not yet known.” A hypothesis such as “the time information may be hard to find” is something to investigate, not an established finding.

Propose a permitted investigation without collecting new personal data

If we later investigate the cause, we could inspect where compatibility information appears in the catalogue or have a teacher organise a permitted comparison of clearer formats.

Do not quietly collect real pupils’ data just because it might help. For this sequence, the supplied fictional records are enough to build and test a paper **process**.

Inspect unsupported conclusions in W2

Use workbook W2 for the connected example “A problem worth solving”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge the AI-is-the-outcome claim

- Use the worked case for A problem worth solving. Proposed explanation: “Using AI is itself the **outcome**”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if **need**ed.

- Answer on your own before discussion. If you are unsure, say which step you **need** help with.

Return to the tray goal

Write a one-sentence project goal and say what **evidence** would count as a failure.

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A strong project begins with a specific **user** and observable **need**.

Use a fresh example to explain **user**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: The first **interview** represents every **user**.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

> **Write a short project brief for the activity problem. Include the users and setting, the supplied evidence, the desired outcome, information needed, one boundary and a simple alternative. Name one claim the ten-request log does not establish**

Practise with less help: Identify claims beyond the evidence

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Investigate a need through a supplied role-play

> Practise neutral questions without inventing personal testimony.

The three fictional evidence cards

Fictional need-finding role-play: use these supplied statements rather than classmates' personal experiences. Requester card: "I had ten minutes and needed a quiet activity. My first choice took twenty minutes, so I chose again." Catalogue card: "Durations and room conditions are listed, but this record does not show why the requester missed them."

Observer card: “Across ten fictional requests, six first choices fit and four did not.”

Ask, distinguish evidence and frame the goal

1. One partner reads the requester card and the other asks neutral questions: What were you trying to do? What happened next? What information would you need to check a compatible choice? Answer only from the cards and mark absent answers unknown. Each learner writes a need statement, a possible explanation labelled hypothesis, a measurable outcome and a simple checklist alternative. Swap roles. Do not claim the role-play establishes real pupils’ preferences or the effectiveness of an unbuilt recommender.
2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Keep the statement and its evidence limits

Case	Prediction	Observed result	What it means
Normal			
Edge			

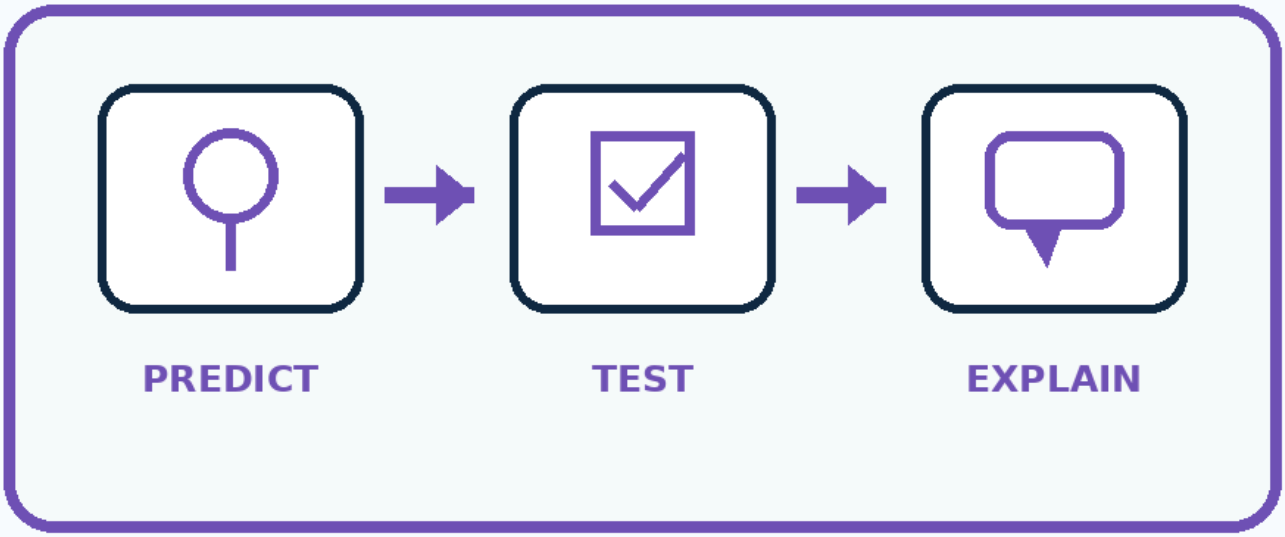
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Need-Finding Interview

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: **Interview** a classmate and write one testable **need** statement.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: frame the equipment-cupboard problem

> Use the supplied box records to write a new bounded brief.

The independent record and task

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do.

The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the **user** difficulty, **evidence**, desired **outcome**, necessary information, an excluded action and a simple alternative.

Identify one cause or benefit that these counts do not establish.

Keep your first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Frame a different problem independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
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Week 29 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Learners often cannot find borrowed classroom books. Define the problem before choosing an AI tool. Suggest one simple solution to compare with AI.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Choose the Classroom Sorter's real **user** and measurable **need**; keep an AI-free solution option open.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Transfer the method to the sorter portfolio

> Keep the connected activity project and earlier sorter context clearly distinguished.

How the two project contexts relate

Two project contexts are available. The connected final sequence develops the fictional activity recommender on paper. This section transfers the same problem-framing method to your ongoing Helpful Classroom Sorter portfolio; it does not change the recommender into a sorter or require a second implementation. Use existing permitted sorter evidence, and label a proposed need as a hypothesis if you lack observations.

This week’s sorter decision

THIS WEEK'S CHECKPOINT Choose the Classroom Sorter's real **user** and measurable **need**; keep an AI-free solution option open.

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

Plan evidence, scope and simpler alternatives

- What real classroom **need** does this serve?

- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Keep personal stories within the design need

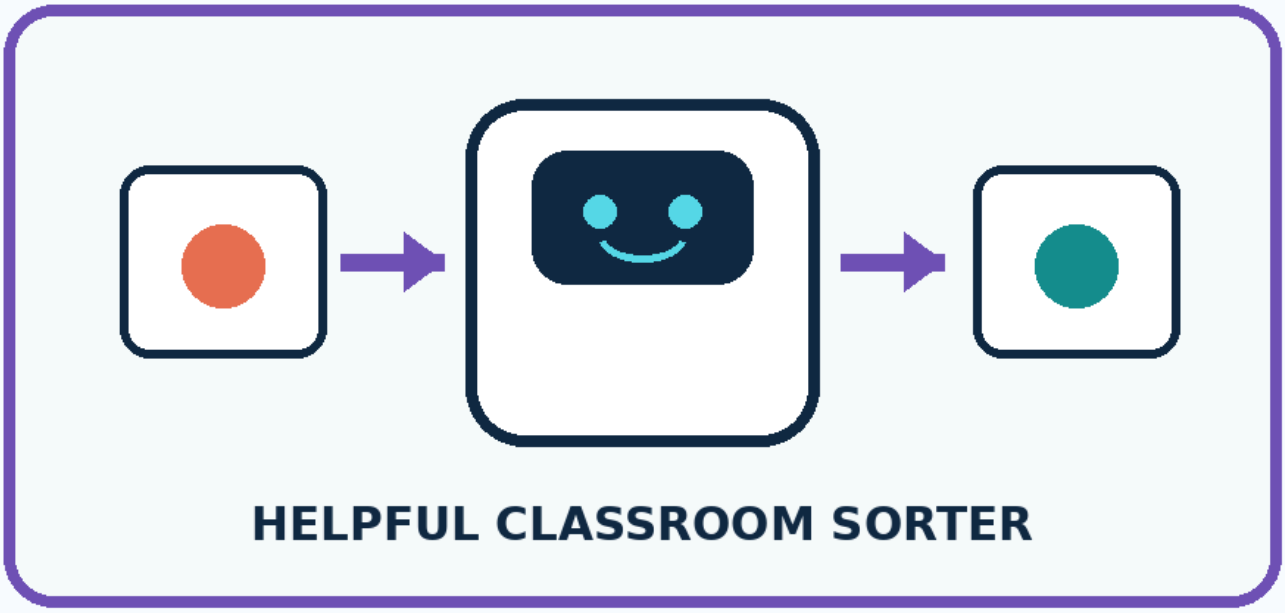
RESPONSIBLE DESIGN Do not collect personal stories unrelated to the design question.

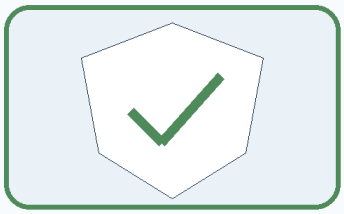
▼ Earlier wording, alternatives and picture transcripts

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain the need before the technology

> Close your notes and name an observable result and a limit of the evidence.

Your exit explanation

- Close your notes. Explain this goal in your own words: Describe a specific **user** difficulty using supplied **evidence** without inventing its cause

My deciding step, evidence and one remaining question

Choose the next changed-case check

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
user	person affected by the problem or solution	
need	observable difficulty or goal	
interview	structured questions used to learn from a person	
success measure	evidence showing whether the need improved	

Terms for the connected explanation

Word	Meaning
User need	Something a person needs to accomplish in a particular situation
Problem statement	A specific description of who faces a difficulty and what that difficulty is
Outcome	The result experienced by the user

Word	Meaning
Constraint	A condition that a proposed solution must respect

Repair a claim using a word

- CLAIM TO REPAIR A popular technology is proof of a real problem.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

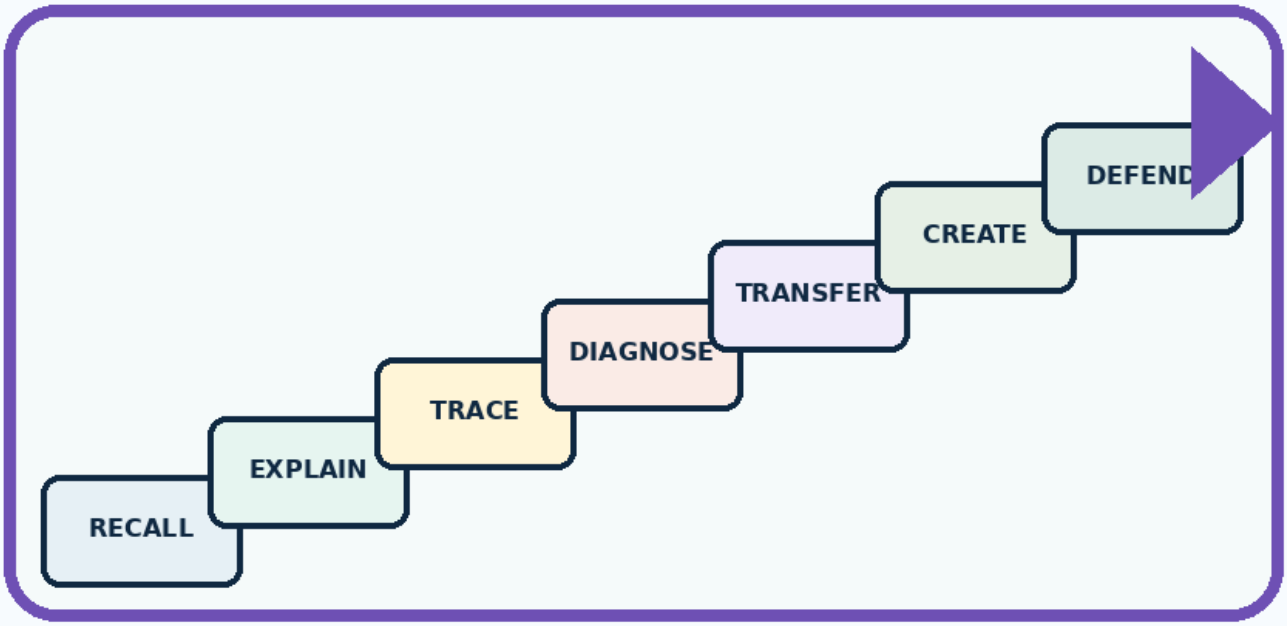
- RECALL Define **user**.
- EXPLAIN A strong project begins with a specific **user** and observable **need**.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: A popular technology is proof of a real problem.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Frame a different problem independently is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

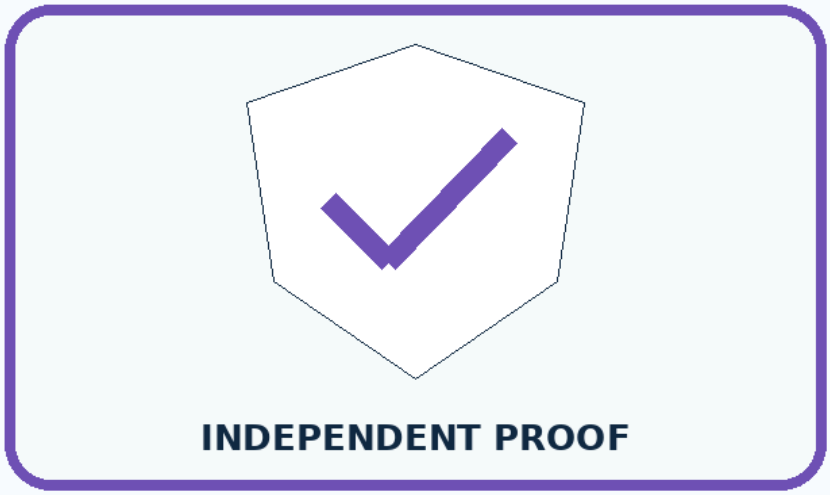
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Frame a different problem independently is the selected independent case.



EXPLANATION CHECK Name the **user**, observed difficulty, bounded outcome, necessary information and simple alternative without inventing a cause.

Independent **evidence**:

SELF-CHECK A strong statement names the **user**, context, observable difficulty, and desired improvement without requiring AI. Protect privacy, avoid leading questions, and distinguish what the **user** said from your inference.

NEXT BRIDGE Carry today's **evidence** habit into: Choose **Input**s and **Output**s.

▼ Extra practice: Solve the need before selecting technology

REFERENCE · SECTION 13

Extra practice: Solve the need before selecting technology

> Try the question before opening its worked explanation.

Read the supplied case and question

Shared markers are often lost because tray labels are unclear.

State the **need** before selecting AI.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Students **need** to return markers correctly and quickly. Test clearer labels before proposing a model.

Evidence or stage	Value or result
Problem	Markers returned to wrong trays
Possible cause	Unclear tray labels
Simple trial	Clearer labels
Evidence	Measure correct and quick returns

▼ Earlier wording, alternatives and picture transcripts

Week 29 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Solve the need before selecting technology

Shared markers are often lost because tray labels are unclear.

State the **need** before selecting AI.

Students **need** to return markers correctly and quickly. Test clearer labels before proposing a model.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Describe a specific user difficulty using supplied evidence without inventing its cause	Explain the deciding step, show a trace or test, and state one limit.
Define a useful outcome and a way to observe it before choosing a solution	Explain the deciding step, show a trace or test, and state one limit.
Choose a small project with clear boundaries, permitted information and a simple alternative	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional class loses time finding the correct craft tray.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional class loses time finding the correct craft tray.
- Which labelled step matches this detail? Use fictional tool requests; do not collect identity or purchase equipment.
- 1. Bound the project
- 2. Observe problem
- 3. Set success test
- 4. Define **need**
- Hint if **need**ed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Bound the project Use fictional tool requests; do not collect identity or purchase equipment. The mismatch record describes a concrete **need**.

A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

Changed-case question

- Changed-case practice: The approved list contains no tray with every requested tool. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. The finder should report no suitable tray rather than guess.
- 3. This one example guarantees correct results for every future **input**.
- Hint if **need**ed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the changed case

After your attempt, compare with the supported explanation: The finder should report no suitable tray rather than guess. Success includes honestly identifying unsatisfied requirements.

Repair and save your work

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system's capability, the correctness of its **output** and permission to take an action. Short practice: Which original proposal is authorised?

Why do the other two **need** separate approval?. Return to this week and explain what you changed.

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Fair for Whom? (ai-1/23). Goal: Compute a correct-out-of-assessed fraction and explain its denominator. Short practice: Which group loses performance under N, and can total improvement erase that fact?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

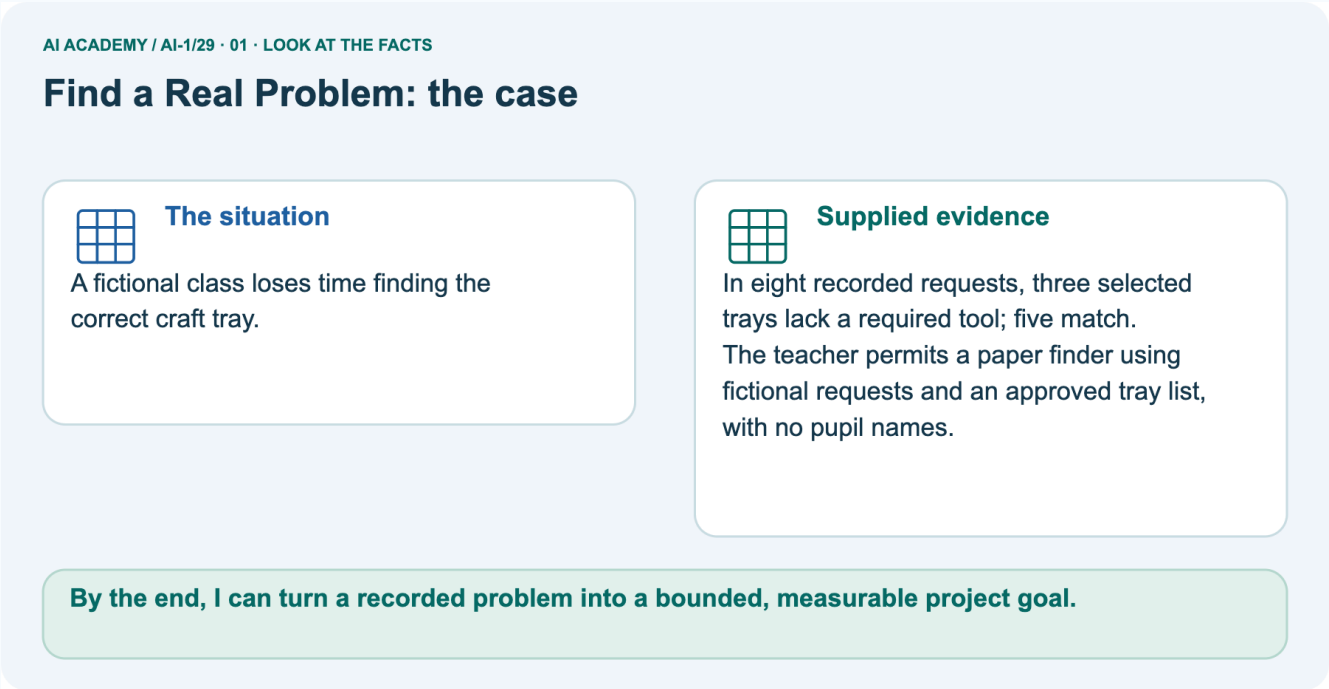
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Find a Real Problem: the case



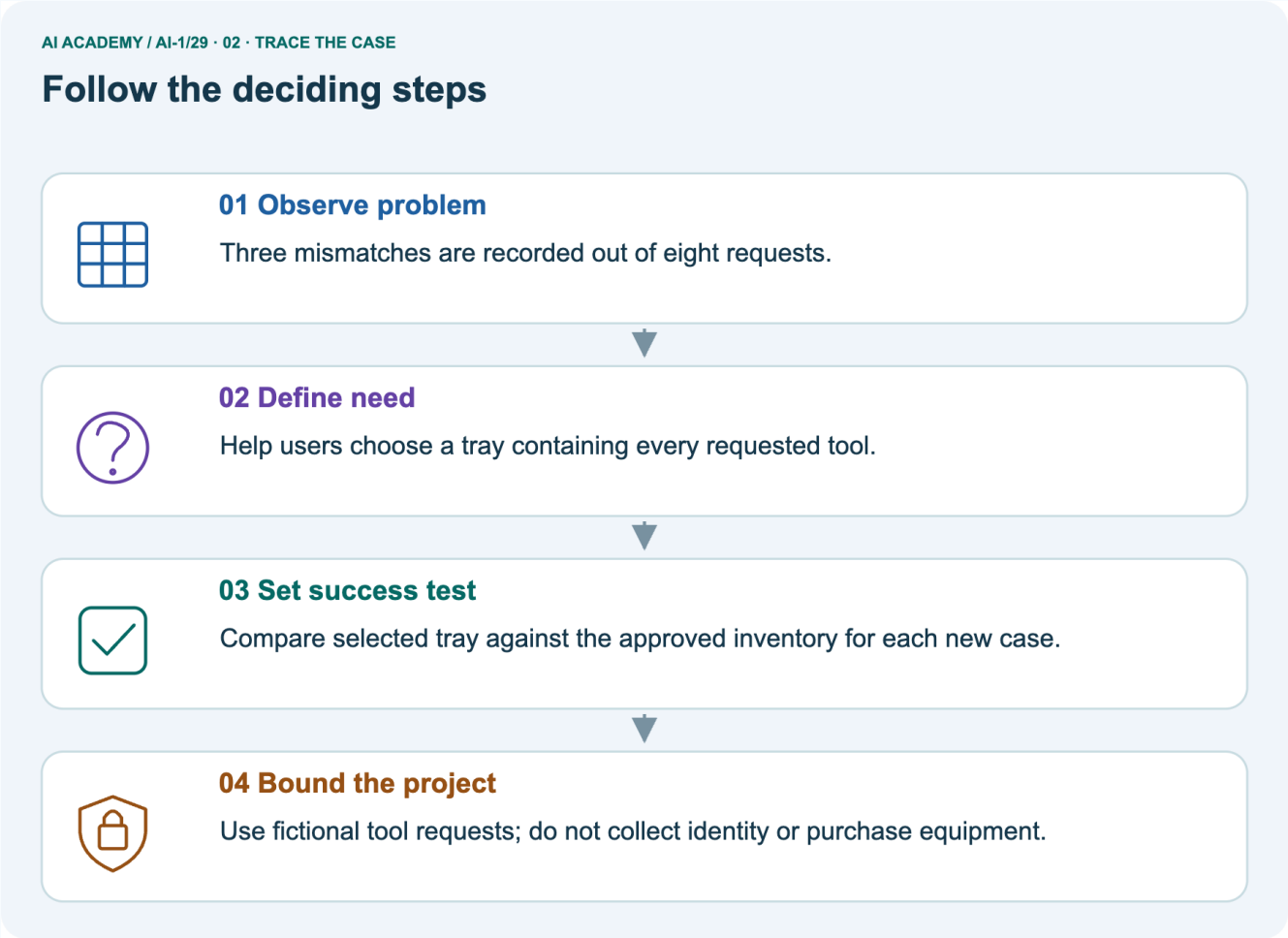
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional class loses time finding the correct craft tray. Supplied facts: In eight recorded requests, three selected trays lack a required tool; five match.

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Observe problem: Three mismatches are recorded out of eight requests. → Define **need**: Help **user**s choose a tray containing every requested tool. → Set success test: Compare selected tray against the approved inventory for each new case. → Bound the project: Use fictional tool requests; do not collect identity or purchase equipment.

Point to each numbered stage and explain which recorded **input** it uses.

03. From an observed difficulty to a testable goal



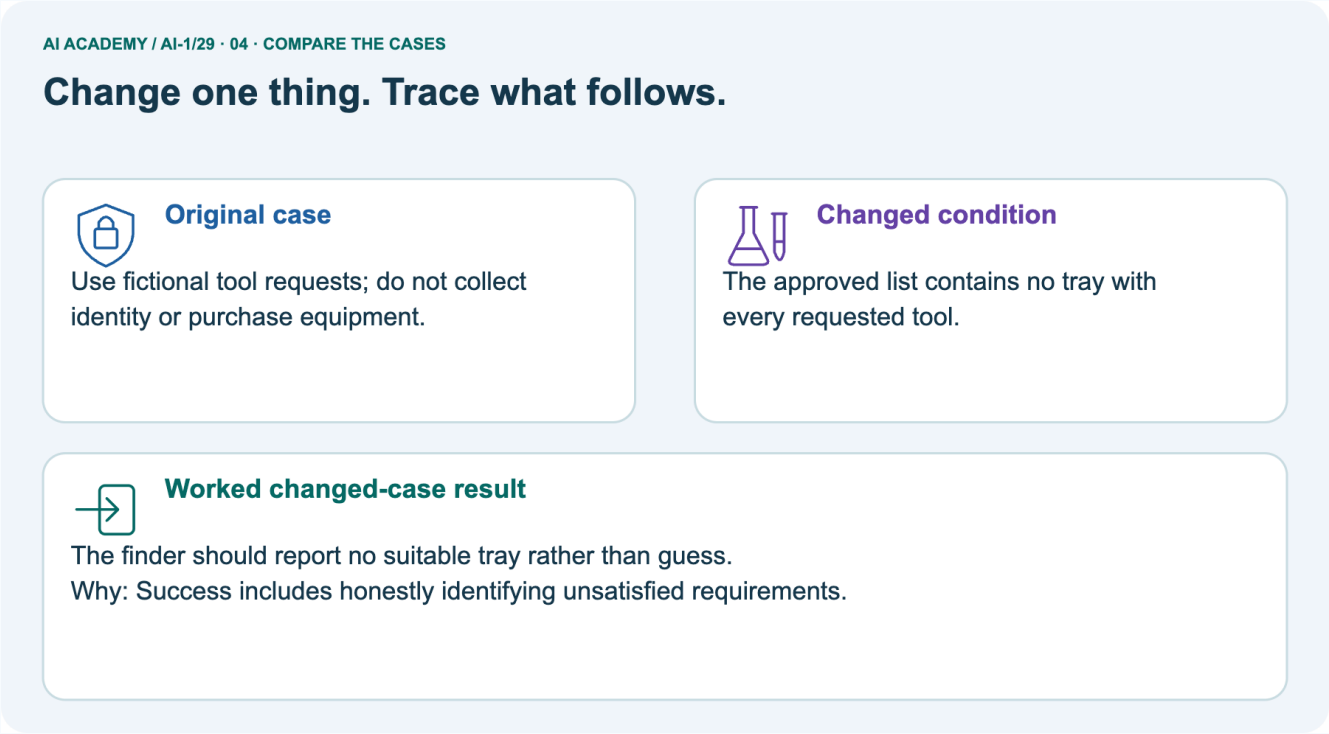
Follow the labelled picture with the matching explanation. Open the vector diagram

From an observed difficulty to a testable goal. 4 of 10 choices mismatch | Some re-selection occurred | Cause is not established. **Need** a compatible choice | Check time and conditions | **User** can still decline.

Paper suggestion **process** | Small permitted prototype | Benefit **need**s a comparison. Measure fit and coverage | Observe useful **outcome**s | Do not hide no-result cases

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



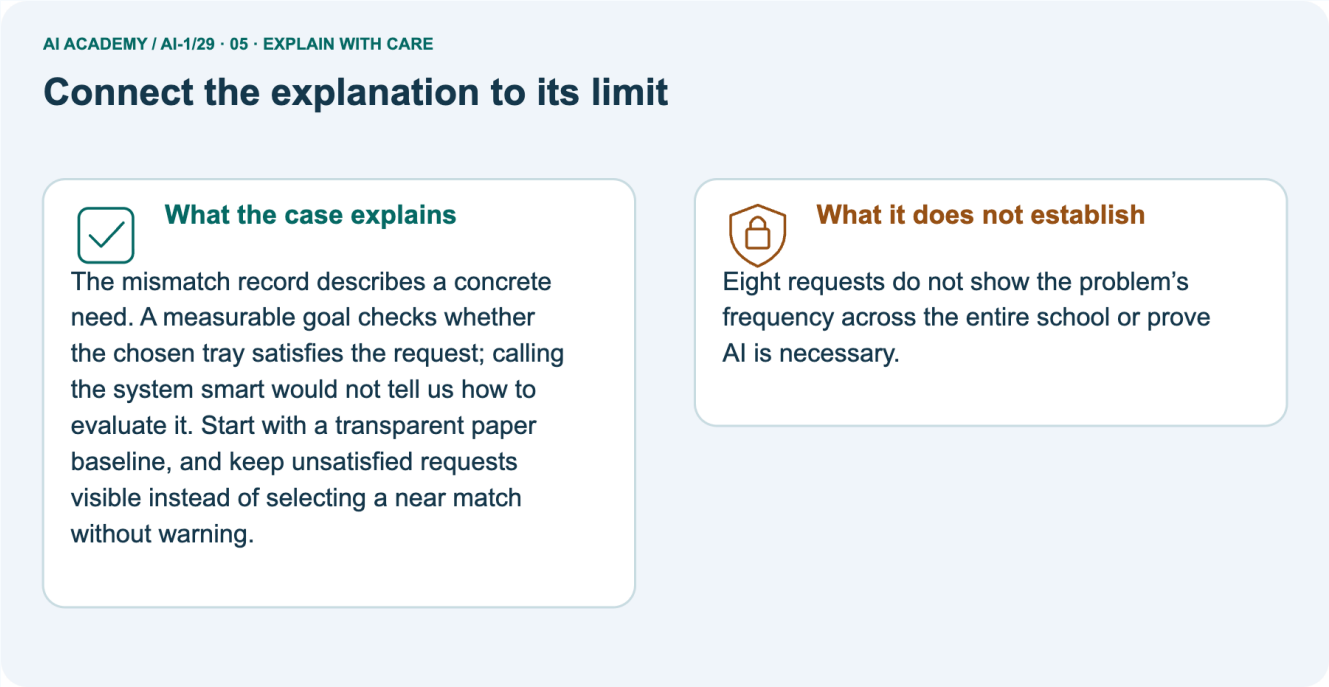
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use fictional tool requests; do not collect identity or purchase equipment.
Changed condition: The approved list contains no tray with every requested tool.

Expected result: The finder should report no suitable tray rather than guess. Reason: Success includes honestly identifying unsatisfied requirements.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

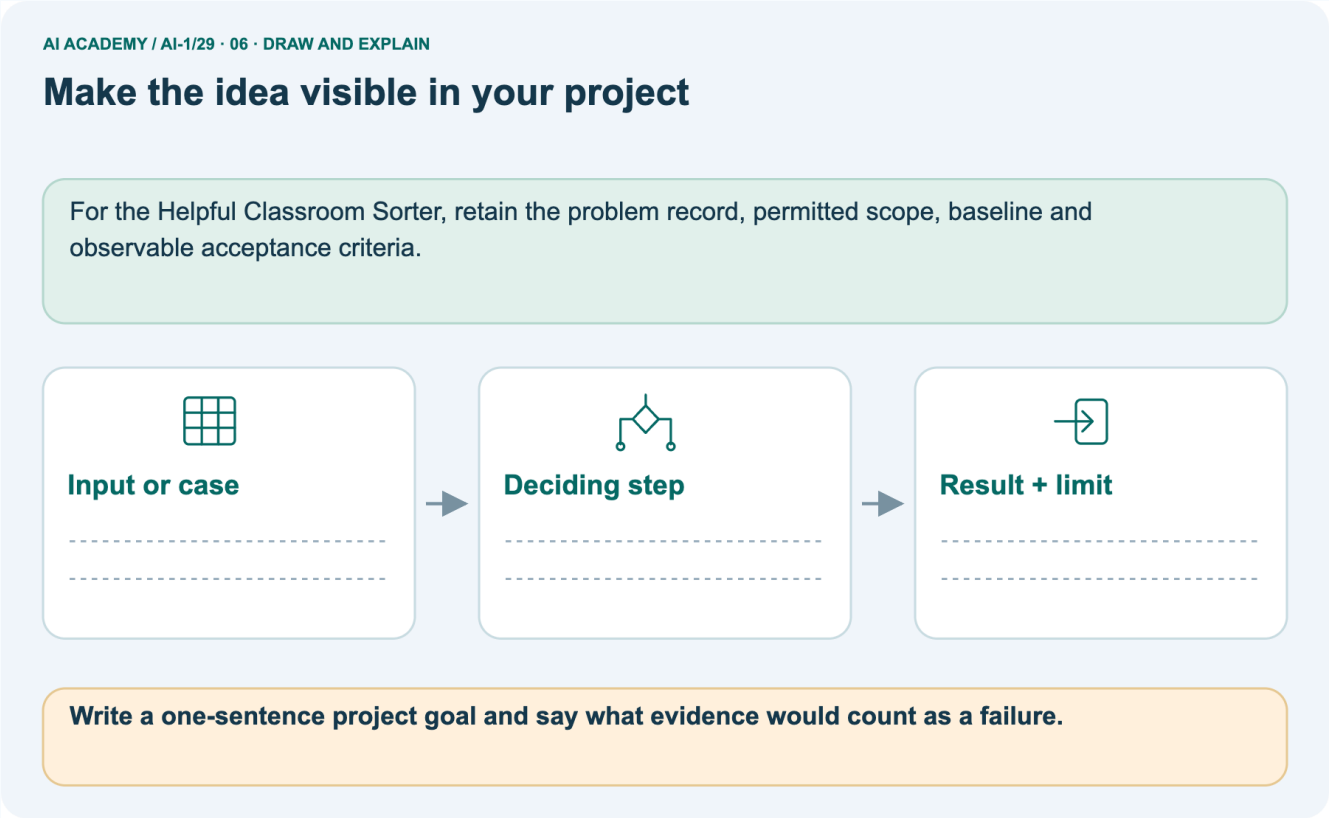
The mismatch record describes a concrete **need**. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it.

Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

Limit: Eight requests do not show the problem's frequency across the entire school or prove AI is necessary.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria. Draw your **input**, deciding step, result and limit.

Supported question: Write a one-sentence project goal and say what **evidence** would count as a failure.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Find a Real Problem: the case

02. Follow the deciding steps

03. From an observed difficulty to a testable goal

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.